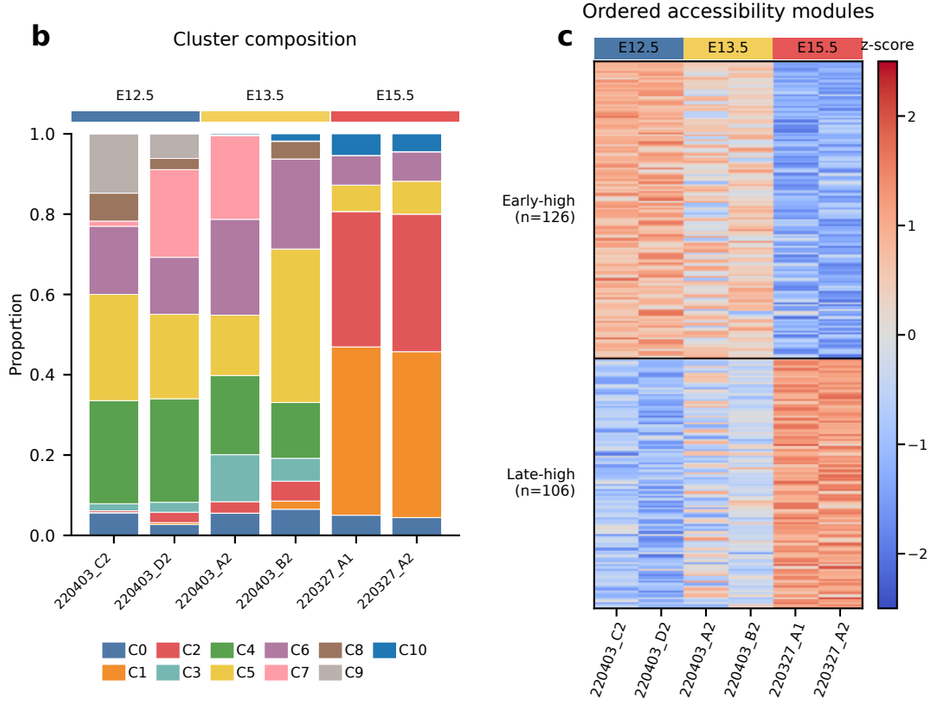
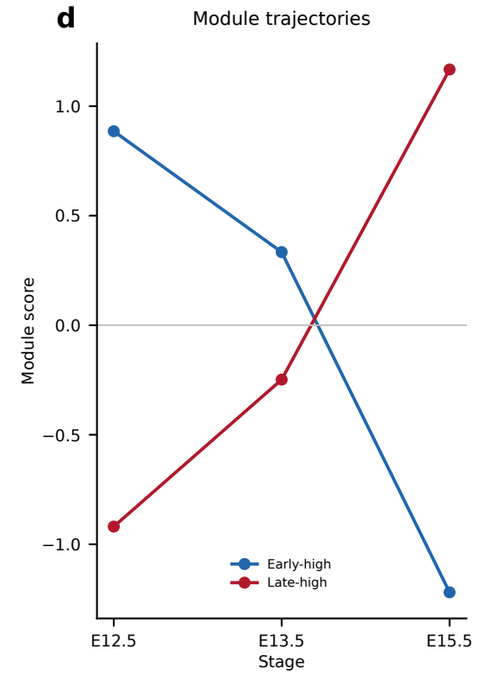
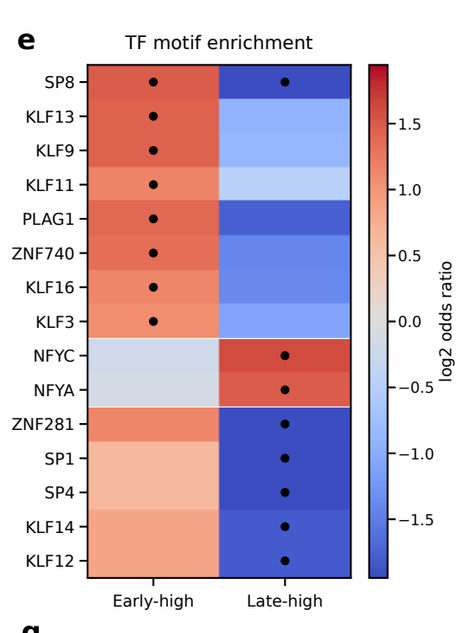
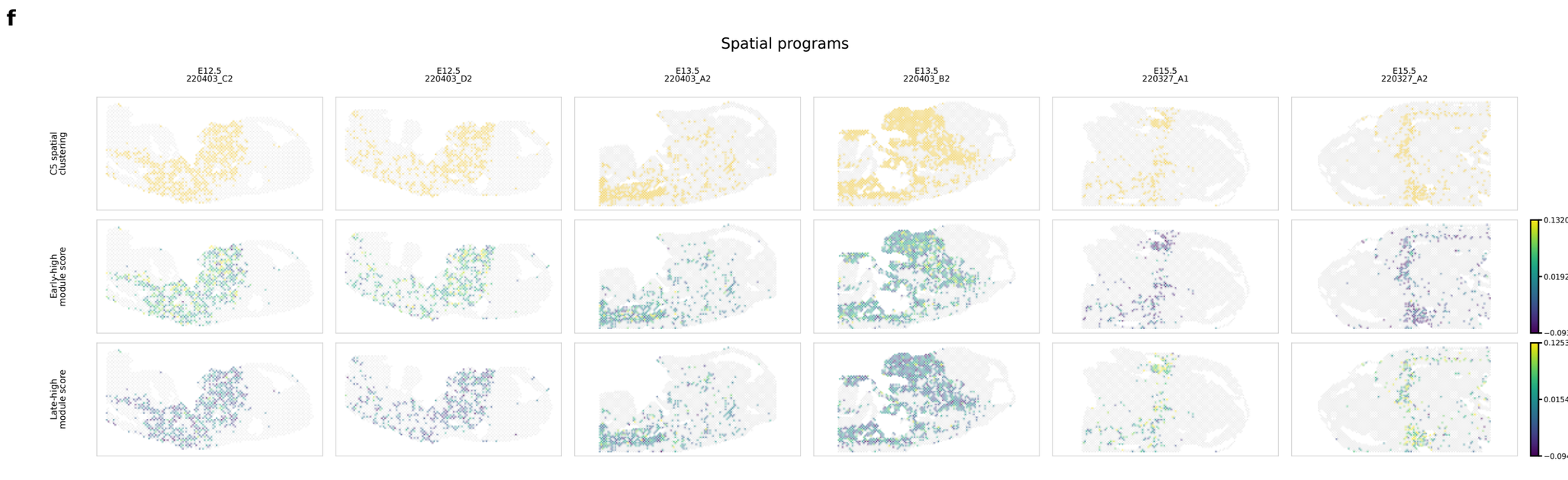

a
E12.5
E13.5
E15.5
220403_C2
220403_D2
220403_A2
220403_B2
220327_A1
220327_A2
Ground truth
SpaCRAFT clusters
SpaCRAFT clusters
C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 C10
Ground-truth cell types
Hindbrain
Internal
Facial_bone
Meningeal_PNS_1
Muscle_heart
Midbrain
Liver
Meningeal_PNS_2
Periventricular
Forebrain
Limb

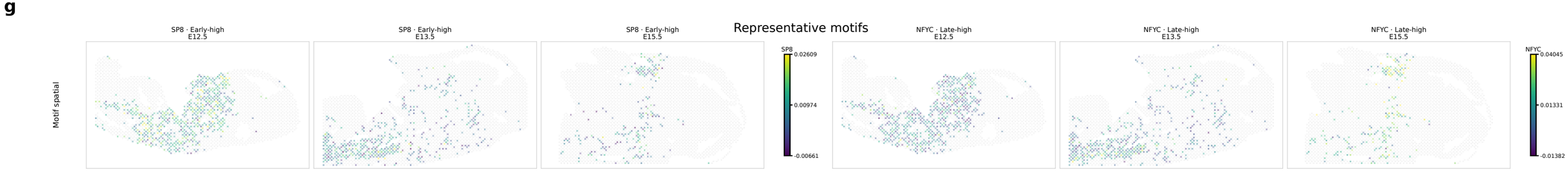

b
Cluster composition
E12.5
E13.5
E15.5
Proportion
1.0
0.8
0.6
0.4
0.2
0.0
220403_C2
220403_D2
220403_A2
220403_B2
220327_A1
220327_A2
C0 C2 C4 C6 C8 C10
C1 C3 C5 C7 C9

c
Ordered accessibility modules
E12.5
E13.5
E15.5
z-score
Early-high
(n=126)
Late-high
(n=106)
2
1
0
−1
−2
220403_C2
220403_D2
220403_A2
220403_B2
220327_A1
220327_A2


d
Module trajectories
Module score
1.0
0.5
0.0
−0.5
−1.0
Early-high
Late-high
E12.5
E13.5
E15.5
Stage


e
TF motif enrichment
SP8
KLF13
KLF9
KLF11
PLAG1
ZNF740
KLF16
KLF3
NFYC
NFYA
ZNF281
SP1
SP4
KLF14
KLF12
Early-high
Late-high
log2 odds ratio
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


f
Spatial programs
E12.5
220403_C2
E12.5
220403_D2
E13.5
220403_A2
E13.5
220403_B2
E15.5
220327_A1
E15.5
220327_A2
C5 spatial clustering
Early-high module score
Late-high module score


g
Representative motifs
SP8 · Early-high
E12.5
SP8 · Early-high
E13.5
SP8 · Early-high
E15.5
NFYC · Late-high
E12.5
NFYC · Late-high
E13.5
NFYC · Late-high
E15.5
Motif spatial
SP8
0.02609
0.00974
-0.00661
NFYC
0.04045
0.01331
-0.01382